



Ahmedabad
University

Object Recognition Enhanced by Action Priming

PSY310: Lab in Psychology

Solo Project

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Github: <https://github.com/YuraIto/Lab-in-Psych/tree/Solo-Project/Participant%20data>

Abstract

Exposure to an object or an image can influence the action related to that object or image. There have been multiple studies on Action Priming, studying this relationship between stimuli representations and action. This particular study aimed to study Action Priming by verifying if observing an action performed by someone else can also enhance the recognition of objects associated with the action, relating action priming to object recognition. A video displaying a motor action performed by either limb (hands or legs) was shown, followed by picture-word matching tasks along with a two-alternative forced choice tasks. Partial pictures of objects are presented followed by two options among which the participant had to select the one they think is the most relevant to the displayed picture. These objects displayed can either be congruent and related to the action in the video, or incongruent and unrelated to the action. The reaction time towards both incongruent and congruent pictures were compared using a paired samples T-test to find that the congruent condition induces a faster response as compared to the incongruent condition. This finding confirms that viewing an action can also prime a person into recognizing the objects related to it faster, as has been found in a few prior studies.

Introduction

Exposure to a stimulus can affect the recognition of another. This is explained using the phenomena of priming which stands for the change in ability to identify an item as a consequence of a prior encounter with the same or related item (Tulving & Schacter, 1990). This provides evidence for the existence of implicit memory, which is responsible for influencing current behavior or choices with respect to past experiences. Implicit memory is an expression of what the individual knows without consciously remembering how, when, or where that knowledge was acquired and priming is an implicit process that makes use of this memory, being completely unrelated to explicit memory (Schacter & Buckner, 1998b).

There have been an extensive number of studies looking at implicit memory with the help of the priming paradigm, which display more accurate identification of recently studied items than of nonstudied ones even without any intentional learning of the studied items taking place (Tulving & Schacter, 1990). Priming takes place in multiple contexts and creates unconscious or implicit associations between multiple items such as objects, emotions, behaviors, or motor movements. Among these associations is one between motor movements and object perception. Views on cognition say that concepts of objects lead to the re-enactment of the motor neural pattern which is usually activated when interacting with the related object, as proven by studies on mirror neurons and canonical neurons. With respect to this, manipulable objects seem to be automatically activating motor neurons (Kalénine et al., 2009). There has been evidence connecting the images of tools to their specific hand movements, showing that pictures of tools commonly associated with specific hand movements elicit activity in regions of the brain that store motor-based information, creating a close link between manipulable objects and their associated actions

(Chao & Martin, 2000). Overall, these studies indicate a possible relationship between actions or context information and the recognition of object representations (Kalénine et al., 2009).

The influence of a motor movement on perceiving a current stimuli is referred to as Action Priming. Past research on effects of action priming on objects has shown that the execution of an action after viewing a manipulable object is affected by whether the object is usually associated with that action or not, also called the congruency (Tucker & Ellis, 2004). Action priming has also been shown to be playing a functional role in object recognition. A previous study looking at object recognition and action priming found higher accuracy for priming video and target object pairs with congruent conditions as compared to incongruent conditions (Helbig et al., 2009b). Even so, there have not been many studies that have looked at the effects of action priming on object recognition, especially through a two-alternative forced choice task. Other studies have also focused on motor movements related to the hands and their associated objects without including leg movements. This lack of literature looking at that aspect is what led to the development of this study.

This study evaluates object recognition with relation to action priming by seeing if the recognition of objects associated with hand and leg actions after being primed with a video showing their respective action is faster as compared to recognition of action-unrelated object images. In simpler terms, **the purpose of this study is to understand whether action representations can facilitate object image recognition. Based on evidence showing action-object associations, the hypothesis for this study is that objects related to an observed action would be recognized faster than unrelated objects.** This could stand to show that perceiving an action would automatically activate mental representations of the objects considered related to the perceived action, strengthening the evidence supporting an interconnection between the motor and object-recognition systems.

Keywords: action priming, reaction time, congruent, incongruent, object recognition, perception

Method

Participants

The study involved the participation of 6 participants. Among all the participants ($n = 6$), 3 were females and the other 3 were males. They had a mean age of 20.5, and are currently residing in the city of Ahmedabad. 5 participants are a part of the Undergraduate division of Ahmedabad University and the one other participant is a part of the PhD program in the same university. They were informed about the usage of their data and were guaranteed their privacy. See *Appendix A*, containing the Ethical Clearance Form, to learn more about the actions taken to ensure participant privacy and ethical guarantee.

Materials

The experiment made use of the v2024.2.4 version of an open software package called PsychoPy for the creation of the experiment through which data was collected from the participants. The trials were conducted on a Windows 10 desktop of model 'Inspiron 3584' with a 1920x1080x60 hertz resolution and a 32 Bits/Pixel colour depth. The dimensions of the screen are 13.54 inches for height and 7.64 inches for width. The response was taken with the help of the laptop's in-built keyboard. There were 4 MP4 File videos used for the experiment, of which three (dancing.mp4, walkingvideo.mp4, and writing.mp4) were obtained from free stock videos and one (handswing.mp4) was recorded through an android phone. The partial object images were obtained by using the snipping tool to snip required aspects of object images found on free stock photo and shopping websites. The images contained partial representations of the following objects: paint brush, crayons, keyboard, frisbee disc, dart, screwdriver, scissors, racket, shoes, slippers, ice skates, skateboard, anklet bells/ghungroo, ballet shoes, and skipping rope. The images used were all of objects that would be closely related to actions carried out either by the hands or by the legs. The data obtained through the trials in the form of Excel sheets was sorted using Google Sheets for each participant. The accumulated data was then tested using JAMOVl, which is an open-source statistical software built on the R programming language.



Fig. 1.1. Partial image of scissors (Left) and partial image of ice skates (Right)

Design

This is a single-factor within-subjects design with one independent variable (Congruency) having two levels (Incongruent and Congruent) and one dependent variable (Reaction Time), where every participant has to respond to both the conditions of congruent and incongruent and their average reaction time to both conditions will be compared. The participants have to respond to 4 images that follow each action video, with two of the images being of the congruent condition and the other two being of the incongruent conditions. Each video is followed by two incongruent and two congruent images. The congruency is determined by whether the partial image is related to the action in the video (congruent) or not (incongruent). Each participant was shown 4 videos in total, where 4 images followed the display of each video, and each image was followed by 2 text options from among which the participant had to choose the

one which represented the image best. In conclusion, there were 4 trials, and each trial had one video followed by 4 images, with each image having two options from which the participant had to choose the option most related to the shown image.

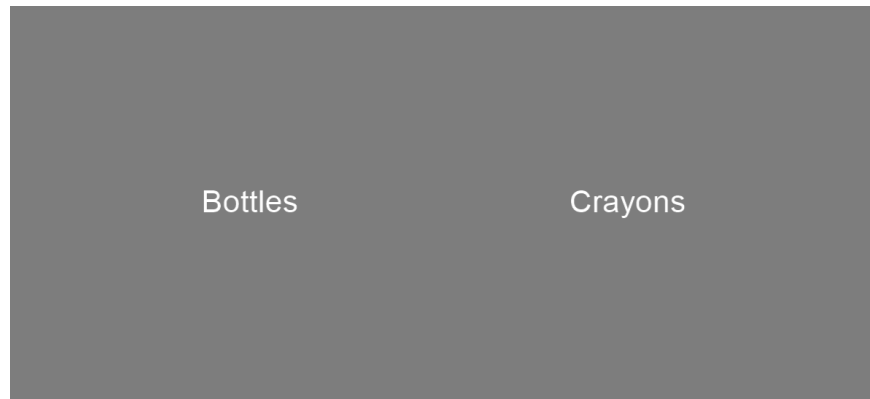


Fig.1.2. Example of text choices displayed on screen after every image

Procedure

At the beginning of the experiment, a grey screen was displayed with the following instructions :

“You will be shown a video followed by partial images of objects. After which you will be given two options.

Select the option that is defining the image shown.

Press 1 to select the left option

Press 0 to select the right option

Press Space to Continue“

The participants were allowed to proceed to the first trial after making sure that they had understood the instructions and were aware of how the collected data would be used.

Each trial began with the presentation of a video, with a width and height of 0.5 units, displaying an action for 15000 milliseconds (15 seconds). 100 milliseconds after the display of a video, an image was presented for 300 milliseconds followed by a blank grey screen that remained for 600 milliseconds till two text options were displayed on screen. The image had a height and width of 0.5 units and the two text options were displayed on the left and right side of the screen with coordinates of (-0.3,0) and (0.3,0) respectively. The participants' choice of a text option was taken with the help of a keyboard key pressing response where they had to press 1 for the left text option and 0 for the right text option. The position of



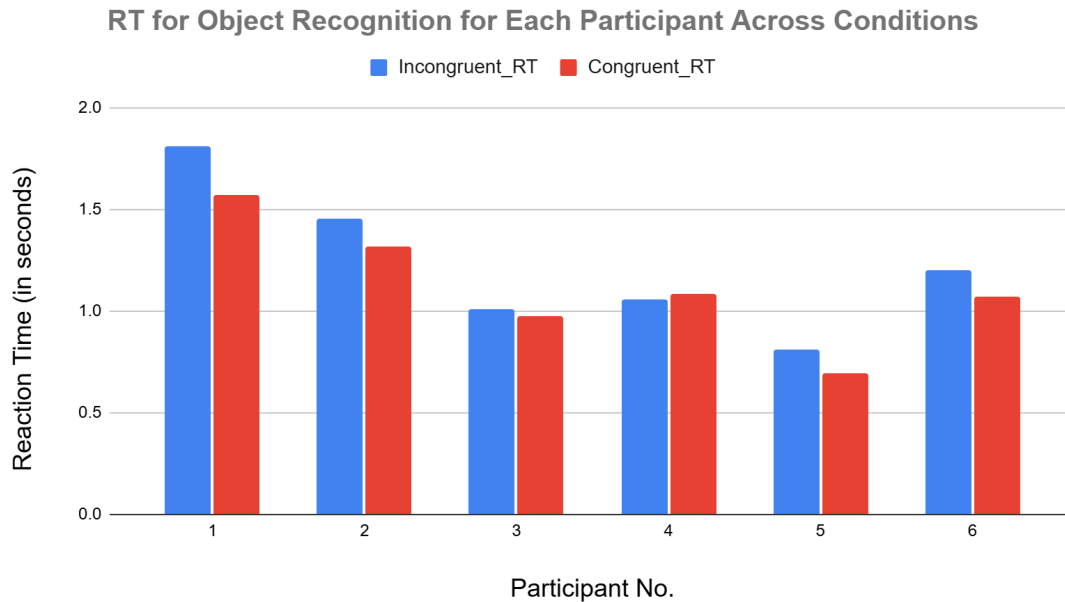
the keys coincide with the spatial location of the options on the screen as 1 is to the left on the keyboard and 0 is to the right, thus the participants had to press 1 with the left hand and 0 with the right. This ensured stimulus-response compatibility, reducing conflict and errors.

There were 4 trials in total, where 2 trials displayed an action done by hands (writing and arm swinging) and 2 displayed actions done by the feet (walking and dancing). Each trial's video was followed by the display of 4 partial images of objects, with 2 images having objects related to the action shown in the beginning of that trial (congruent) and the other 2 showing objects unrelated to that action (incongruent). The total number of trials each participant among the 6 participants responded to were thus 16, giving data from a total of 96 trials. From among these, the 8 trials in which the participants gave incorrect responses were removed to ensure accuracy, reducing it to a total of 88 trials. This shows that the participants had an overall mean accuracy of 92%. The images in each trial were presented in a random order to increase validity by making sure there is no carryover effect. The Reaction Time, which is defined as the time interval between the display of the image and the pressing of a key, was sorted and used for further calculation.

A paired samples T-test was conducted using the average reaction time towards both the congruent and incongruent components. This aimed to understand whether there is a significant difference between the average reaction time towards the congruent stimuli or images and the average reaction time towards the presentation of the incongruent stimuli. According to the proposed hypothesis, the average reaction time towards the images which show objects related to the action shown in the trial should be lower than the reaction time towards images with objects which are unrelated to the shown action. This indicates faster object recognition when the object is associated with the displayed motor action, meaning that viewing a motor action can induce faster object recognition for objects related to that action. Using a within-subjects design allowed direct comparison of reaction times and reduced errors that could be caused due to participant differences in perception and reaction speed.

Results

The observed reaction times for all participants across both the conditions were obtained to be as follows:



A Paired Samples T-test determined the presence of a difference in the dependent variable (Reaction Time) between the two levels (Congruent and Incongruent) of the independent variable (Congruency). The analysis shows that there was a significant difference in reaction time between the congruent ($M = 1.12$, $SD = 0.300$) and incongruent ($M = 1.22$, $SD = 0.359$) conditions, $t(5) = 2.82$, $p = 0.037$, $d = 1.15$.

Paired Samples T-Test

			statistic	df	p	Mean difference	SE difference	Effect Size	
Incongruent_RT	Congruent_RT	Student's t	2.82	5.00	0.037	0.104	0.0370	Cohen's d	1.15

Note. $H_a: \mu_{\text{Measure 1}} - \mu_{\text{Measure 2}} \neq 0$

Fig.2. Paired Samples T-Test showing a significant difference between the two levels of the IV, $p = 0.037$,

Descriptives

	N	Mean	Median	SD	SE
Incongruent_RT	6	1.22	1.13	0.359	0.147
Congruent_RT	6	1.12	1.08	0.300	0.122

Fig.3. *Descriptive statistics of the two different conditions.* The values mentioned here are the number of participants (N), the Mean (M), the Standard Deviation (SD), the Median, and the Standard Error (SE).

This suggests that the condition of congruency had a significant effect on the Reaction Time, indicating that object recognition is affected by the condition of congruency with the displayed motor action. Taking into consideration the observed mean reaction times for both the conditions (Incongruent_RT M = 1.22; Congruent_RT M = 1.12, we can support the hypothesis that object recognition is faster for objects associated with the displayed motor action as compared to that for objects unrelated to the action. Thus, there is not enough evidence to prove the null hypothesis that there is no difference between the reaction times towards both the congruent and incongruent conditions here.

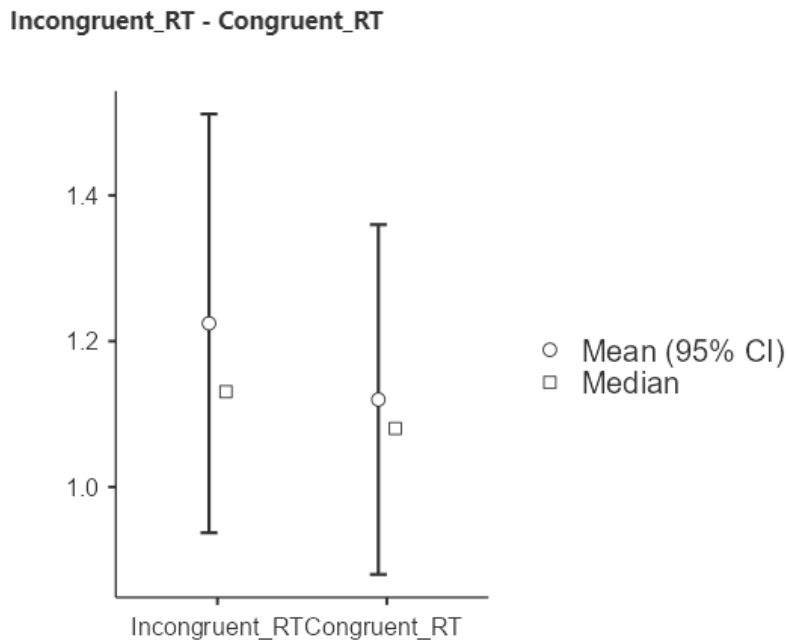


Fig.4. *Descriptive plot showing change in reaction time (DV) with change in congruency (IV).* The X-axis stands for the two levels of the IV and the Y-axis displays the DV (Reaction time) in seconds.

Discussion

This study had the main aim of providing evidence to support the hypothesis that recognition for objects related to a displayed action would be faster as compared to recognition of unrelated objects. The findings have supported this initial proposed hypothesis through a faster average reaction time towards the congruent condition in which the participants have to recognize the partial image of a displayed object which is usually associated with the action video shown in the beginning of that trial.

This finding contributes to the understanding of how objects representations in our memory are activated implicitly upon perceiving a motor action. **The faster reaction towards the congruent condition could suggest that perceiving an action taking place activates a series of rapid semantic activations that also includes knowledge of associated objects, making it such that information on objects related to the action are readily available for retrieval by the brain. This agrees with the findings from Helbing et al.'s paper where it is shown that observing an action sequence can prime the recognition of a subsequently presented manipulable object that is usually associated with a similar action (Helbig et al., 2009).** This could also further support the presence of a functional relationship between the systems in the human brain responsible for object recognition and object-involving actions. The objects used in this experiment however, unlike the previous studies, are not restricted to tools but also involve objects (like shoes) which are usually heavily associated with an action (such as walking). This could suggest that these object-action associations formed in the brain are not related to just the objects which are used using the action, but also objects which may be correlated to the action although not directly used by it. There is thus a possibility of multiple levels of object-action associations in the brain which go further than the direct action-tool correlations.

The findings from this study can help understand the extent to which past experiences encoded in the sensorimotor areas of the brain can influence current perceptual processes like object recognition, allowing a more efficient processing of the present environment. It can also give insight into possible faults in the current perception of certain patients such as in the case of trauma. By involving two domains in this study, namely the motor and perceptual domains, this study proves the possibility of priming occurring across multiple domains, supporting the theories that put forward that knowledge is organized as sensorimotor experiences.

The practical implications for these findings could affect the sectors like education, rehabilitation, marketing, etc. Action videos can be used to facilitate learning object recognition in developmental disorders or help break traumatic associations formed between an action and an object. This concept can also be applied in schools to enhance the learning of object recognition in children. Finally, the limitation of this study is that it uses a small sample size and thus future experiments working on a similar study could increase their number of participants to increase accuracy.



References

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Appendix A

Ethical Clearance Application Form

Details of Project

Sr. No.	Title	Particulars
1	Title of research project	Object Recognition Enhanced by Action Priming
2	Anticipated duration of the project/trial	1-2 weeks
3	List and details of location(s) where project will be conducted [Include all the collaboration with institute/hospital/practitioner]	The experiments will be performed inside the psychology lab at the School of Arts and Sciences, Ahmedabad University
4	Source of funding & financial allocation for the project / trial	This research is not funded by any agency.
5	Will the research proposal be submitted to any ministry screening committee for international collaboration?	No
6	Brief summary of research proposal	This study examines the influence of action priming via the display of an action video on the recognition of an object usually associated with the action displayed in the video.
7	Research objectives	1) To understand the impact of action priming on object recognition. 2) To understand if the brain's motor domains interact with the perceptual domain.

8	Methodology and Technical approach of the proposal	The study follows a quantitative and experimental methodology using one within-subject experiment. The variables of interest are manipulated in a controlled environment. Participants will complete one laboratory session lasting approximately 3 minutes, including setup, task performance, and debriefing. Stimuli will be presented using PsychoPy on a Windows 10 PC. Participant responses will be collected through key presses on the keyboard. Participants have to select from among two options via key presses. The Reaction Time and Accuracy will be measured through PsychoPy data.
9	Usefulness of the project / trial	The proposed study attempts to explain whether sensorimotor experiences such as viewing an action can enhance object recognition. The findings may be able to help in the creation of AI algorithms. It would provide concrete findings that support the fact that priming is an implicit process that may involve multiple cognitive domains.

Details of Investigator(s)

Sr. No.	Title	Particulars
1	Name of the Principal Investigator (PI) with Qualification, Designation, School, Email Address and, Phone Contact	Rashmita Ganguly BA (Hons) Student, School of Arts and Sciences, Ahmedabad University rashmita.g@ahduni.edu.in +91 7044506033
2	Name of the Co-Investigator(s) Co-PI) with Qualification, Designation, School, Email Address and, Phone Contact	NA
3	Name of Supervisor (In case of student involvement) Qualification, Designation, School, Email Address and, Phone Contact	Professor Nithin George, PhD Assistant Professor School of Arts and Sciences, Ahmedabad University nithin.george@ahduni.edu.in 8488865331
4	Name(s) of student involved with programme name and semester	Rashmita Ganguly BA (Hons) Student, School of Arts and Sciences, Semester 5
5	Name(s) and status of other investigators including students and professional collaborators	NA

Details of ethical consideration for research/project

Sr. No.	Title	Particulars
1	Is approval by other Ethics Committee required? If 'Yes' mention the details.	Yes
2	Type of review requested	Exempted Review/ Expedited Review / Full Committee Review
3	Justify why a human study is needed to answer the research questions.	The proposed study will be using data involving reaction speed towards a presented stimuli and thus needs human participants to understand the perceptual process of object recognition that takes place in the human brain with relation to the implicit process of action priming.
4	Participant recruitment process and eligibility criteria	Participants will be recruited through word of mouth and availability. There is no particular eligibility criteria other than the fact that the participants should not have any dysfunction in their perceptual abilities
5	Number of participants required with necessary justification.	As per the requirement of around 100 trials worth of data, there is a need for at least 6 participants to obtain a total of 96 trials worth of data.
6	Justify inclusion and exclusion of vulnerable population	Individuals with visual deficits are excluded because the experimental task is screen-based and requires normal visual acuity to perform. Individuals with a dysfunction in object recognition are excluded because the experiment requires them to recognize objects based on their partial representations
7	Procedure for seeking and obtaining informed consent with a sample of the patient/participant information sheet and informed consent forms in English and local languages. [AV recording if needed]	The participants will be asked for consent orally at the start of the experiment after being informed of the objectives of the study, along with any potential risks and benefits it entails. They will also be informed

	Informed consent for stored samples	about how their data will be used in detail.
8	Plan for statistical analysis of the study	The data obtained from the experiment (reaction time and accuracy of the responses) will be analyzed using a Paired Sample T-test. Statistical Analysis will be performed using JAMOV, version 2.6.23.
7	Explain the plans to maintain confidentiality of records / data of the study participants	All participant data will be number-coded, and the file will not be tagged with the participant's name. The data will be stored in a password-protected location. The participant's name or any other sensitive information will not be made public.
8	Explain all anticipated risk (adverse events, discomfort) or injury that may be caused to the participant	There is no risk or benefit for the participant.
9	Efforts taken to minimize the risk or injury	The study involves tasks that are not likely to cause injury. Participants will be explicitly informed that they can walk out of the experiment at any time.
10	Whether 'wage compensation' for the research subjects will be provided? [Compensation/ Reimbursement of incidental expenses and management of research related injury/illness]	Participants will receive no compensation for their participation.
11	Expected 'benefits' to volunteer / community	No benefit to the volunteer.
12	Account of storage and maintenance of all the human material or data obtained from the study	The data will be stored in a password-protected private server.
13	Explain the plans of publication of results (positive or negative) while maintaining confidentiality of personal information/identity	Results (positive or negative) will be published in a peer-reviewed journal. During the publication of the findings, no personal information of the participant will be reported.
14	Specific ethical issues, as identified by the investigating team	None.
15	List of documents enclosed for ethical review	1) Detailed project proposal 2) Sample consent form
16	Disclose Conflict of Interest, if any.	None.



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